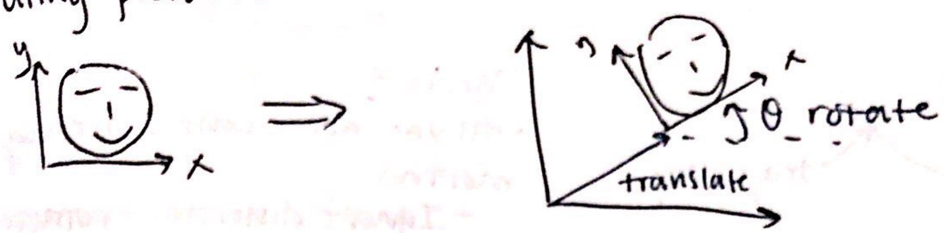
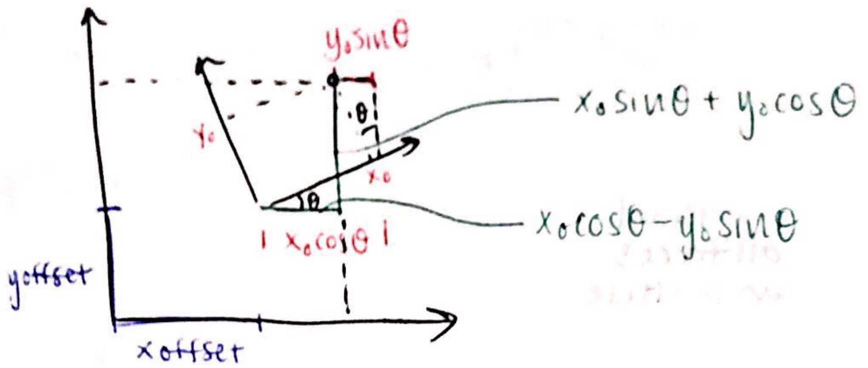
MONDAY AUGUST 27th 2018 LECTURE 2

Rotating pictures:



Picture made of points:



$$\begin{bmatrix} x_{new} \\ y_{new} \end{bmatrix} = \begin{bmatrix} x_{offset} \\ y_{offset} \end{bmatrix} + \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}_{R} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

[new pic] = [offset] + R · [old pic]   Motion = offset + rotation

$$[\text{new array}] = \begin{bmatrix} x_{off} \\ y_{off} \end{bmatrix} + R \cdot \left[\text{point}_1 \mid \text{point}_2 \mid \ldots \mid \ldots \right]$$

· Each point makes up columns of a matrix